

A Multilingual Comparative Evaluation of Zero-Shot Voice Cloning Architectures for English and Hindi Speech Synthesis

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Abstract—Zero-shot voice cloning has improved rapidly, but the way these systems are evaluated has not kept up. Published benchmarks usually cover a single high-resource language, report one quality metric, and rarely compare architecturally different systems under the same conditions. Hindi in particular tends to appear only as one entry in a multilingual training mixture rather than as the subject of a dedicated study. We compare three recent systems that differ in how speaker identity enters the model: Orpheus-3B, which prepends the reference audio as a token prefix to an LLM backbone; VoxCPM2, which cross-attends to a fixed speaker prototype; and VibeVoice, a diffusion decoder guided by a contrastive speaker encoder. All three receive the same 25.5-second reference recording and the same twelve English and Hindi utterances, and all outputs are scored by the same three pipelines: speaker-embedding cosine similarity, Whisper-based word and character error rates, and UTMOS naturalness prediction. None of the systems is best on every metric. VoxCPM2 preserves speaker identity and pitch register most accurately and produces the most intelligible speech in both languages. VibeVoice scores highest on naturalness and degrades the least from English to Hindi, but is unreliable on exact content such as digits and alphanumeric codes. Orpheus-3B is weakest on identity and pitch. Within each system, the English and Hindi speaker-similarity means differ by at most 0.005 even though intelligibility drops sharply, which suggests that a large part of the measured Hindi error comes from the ASR and romanization steps of the scoring pipeline rather than from the synthesis itself. On this protocol, persistent cross-attention to a speaker prototype was a more reliable identity mechanism than an autoregressive prefix.

Index Terms—zero-shot voice cloning, text-to-speech, speaker verification, multilingual speech synthesis, Hindi TTS, neural codec language models

I. INTRODUCTION

Text-to-speech synthesis has progressed from rule-based concatenative methods to neural models whose output is often difficult to distinguish from recorded speech. Zero-shot voice cloning pushes this further: given a short reference recording, typically 20 to 30 seconds of natural speech, the model must reproduce that speaker’s voice on arbitrary new text without any speaker-specific fine-tuning. This requires more than accurate signal reconstruction. To carry a voice onto sentences the reference speaker never actually said, the

model has to separate speaker-characteristic properties (timbre, prosodic habits, vocal tract resonance) from the linguistic content. Orpheus-3B [1], VoxCPM2 [2], and VibeVoice [3] approach that separation in structurally different ways, and the aim of this paper is to measure how those design choices behave under one controlled comparison.

Neural codec language models [4], [5], flow-matching decoders [6], and diffusion-based vocoders have produced zero-shot systems whose mean opinion scores approach ground-truth recordings, at least in controlled English evaluations. Evaluation practice has been less consistent. Most published benchmarks use clean studio-quality reference audio, target one high-resource language, and report MOS under conditions that may not reflect deployment. The situation is worse in multilingual settings. Hindi is spoken by more than 600 million people, but its phonological properties, including retroflex consonants, contrastive vowel nasalization, and an ambiguous Devanagari-to-phoneme mapping, make both acoustic modeling and ASR-based intelligibility scoring harder than the English case that most benchmarks are built around. A model with a low English word error rate (WER) can still fail badly on Hindi if its architecture was never trained on phonologically diverse corpora.

Speaker identity preservation adds a second difficulty. A 20–30 second clip carries only so much speaker-specific information, and the model has to extract a stable representation from that limited evidence and hold onto it across target sentences of varying length and phonetic content. On the measurement side, cosine similarity between embeddings from an independent speaker encoder (a GE2E d-vector network, for instance) gives a reproducible proxy for perceptual identity, although it does not capture every dimension of how listeners recognize a voice. Naturalness is just as hard to measure at scale. Neural MOS predictors such as UTMOS [10] avoid the cost of human listening panels, but how well they transfer across languages remains an open question.

In this work we evaluate three contemporary zero-shot voice cloning systems: **Orpheus-3B** [1], a large language model-based TTS system adapted for zero-shot speaker conditioning;

VoxCPM2 [2], a codec-based generative model trained on diverse multilingual speech; and **VibeVoice** [3], a diffusion-based synthesis pipeline with a contrastive speaker encoder. Each system receives an identical 20-to-30 second reference clip and the same target texts. Outputs are scored on three complementary axes: speaker similarity (cosine distance in speaker embedding space), intelligibility (Word Error Rate for English and Character Error Rate for Hindi, both transcribed with Whisper [11]), and acoustic naturalness (UTMOS). We use CER rather than WER for Hindi deliberately: Whisper’s Hindi output shows transliteration inconsistencies at word boundaries, and character-level comparison separates phoneme fidelity from tokenization artifacts.

The comparison matters beyond benchmarking because each architecture makes different assumptions about the speaker conditioning mechanism, the decoder, and the training regime. Orpheus-3B conditions generation autoregressively through a speaker-prefixed token sequence; VoxCPM2 uses cross-attention to a speaker prototype inside a codec language model; VibeVoice guides a diffusion process with a contrastive speaker embedding. Knowing which of these holds identity more stably from a short reference, and whether that stability carries across scripts, is useful both for choosing a model to deploy and for deciding which architectural directions are worth pursuing in multilingual TTS research.

The rest of the paper is organized as follows. Section II reviews related work on neural TTS, zero-shot speaker adaptation, and multilingual synthesis. Section III describes the experimental setup, including the corpus, preprocessing, and metric implementations. Section IV presents the quantitative results, Section V analyzes the observed trade-offs and their architectural correlates, and Section VI summarizes the findings and outlines future work.

II. RELATED WORK

A. The Evolution of Zero-Shot Voice Cloning

Early neural TTS systems such as Tacotron [12] and Tacotron-2 [13] treated speaker identity as fixed at training time. They raised naturalness considerably, but they could only produce the voices they had been trained on, and supporting a new speaker meant retraining. Cloning an unseen voice requires the model to separate speaker identity from linguistic content in some explicit or implicit way.

Jia et al. [14] showed that embeddings from a pretrained speaker-verification network could condition a synthesizer on a new voice from only a few seconds of audio. Arik et al. [15] examined the trade-off inherent in this approach: an external speaker encoder allows adaptation without retraining, while fine-tuning the synthesizer itself gives better quality for that one speaker at much higher cost.

Later systems such as VITS [16] and FastSpeech-2 [17] improved quality and inference speed but kept the same basic separation. YourTTS [18] extended it by training across many speakers and languages simultaneously, which improved zero-shot behavior, though quality remained limited mostly by

what the speaker encoder could represent rather than by the synthesizer itself.

A larger shift came from reframing TTS as language modeling over discrete audio tokens. VALL-E [4] showed that an autoregressive transformer over neural codec tokens can clone a voice through in-context prompting alone: the reference clip is placed before the target text, and the continuation inherits the speaker without any explicit embedding bottleneck. This built on the hierarchical token modeling that AudioLM [5] had introduced for general audio.

The systems that followed mostly kept the token representation and changed the decoding procedure. Voicebox [6] replaced autoregression with flow matching over masked spans, allowing faster non-causal infilling. NaturalSpeech 3 [19] factorized content, prosody, and timbre into separate discrete subspaces. TorToise [20] showed that simply scaling a fairly standard diffusion-decoder pipeline closes much of the remaining naturalness gap without architectural novelty.

The three systems evaluated here sit at different points on this trajectory. Orpheus-3B [1] follows the VALL-E approach but builds directly on a pretrained LLM backbone, with the reference speaker supplied as a token prefix, so cloning depends entirely on the language model’s in-context learning. VoxCPM2 [2] keeps the codec-language-model design but replaces the prefix with cross-attention to a fixed speaker prototype, an intermediate point between pure prompting and an explicit embedding. VibeVoice [3] departs from autoregressive codec modeling altogether and uses a diffusion decoder conditioned on a contrastively trained speaker encoder, which places it closer to representation-learning systems such as StyleTTS 2 [21].

That three recent production-grade models disagree this much about how identity should be injected suggests the field has not settled on a conditioning mechanism. This is the main reason a side-by-side evaluation seemed worthwhile to us: results reported separately, on different data and different metrics, do not answer the question.

B. Cross-Lingual and Multilingual Generalization

Most of the zero-shot cloning literature is anchored to datasets and evaluation protocols built around a handful of high-resource, mostly Latin-script languages. YourTTS [18] was among the first to report cross-lingual cloning and found that transferring a speaker into a typologically distant language costs more naturalness than transfer within a related family. XTTS [22] scaled multilingual pretraining substantially further, but its language pool still concentrates on Indo-European and East Asian languages, with lower-resource languages evaluated as secondary conditions rather than as the subject of the study. VALL-E X [23] addressed the cross-lingual case directly by training a shared codec language model on English and Mandarin, and reported a measurable loss of speaker similarity when an embedding learned on one phonetic inventory has to generalize to another.

Hindi is further from English than the pairs typically studied. It has retroflex stops that English lacks, contrastive vowel

nasalization, and schwa-deletion rules that make a Devanagari grapheme sequence an unreliable guide to the phonemes actually produced. Extending the VALL-E X observation to Hindi is therefore not just a matter of adding training data in a new language; it tests whether the conditioning mechanism itself is language-invariant.

Indic-language speech technology has largely developed as a separate resource-building effort rather than in dialogue with the cloning literature above. AI4Bharat’s work at national-language scale, spanning ASR [24], standardized recognition benchmarking [25], and large multi-task speech corpora [26], is oriented toward recognition rather than generation, and earlier end-to-end Indian-language TTS [27] depended on hours of curated single-speaker recordings, the opposite of the 20–30 second zero-shot setting adopted here. Zero-shot cloning results specific to Hindi remain sparse. Models often list Hindi among many training languages, but they rarely report Hindi speaker similarity, intelligibility, and naturalness with the same rigor as their English numbers, so it is hard to tell whether claimed multilingual competence reflects genuine phonological robustness or simply the composition of the pretraining corpus.

C. The Transition to Automated Evaluation Pipelines

For most of neural TTS history, the mean opinion score (MOS) listening test was the accepted arbiter of quality, but crowdsourced MOS collection has known reliability problems. Ribeiro et al. [28] documented substantial rater variance in crowdsourced studies, and Cooper et al. [29] showed that a MOS predictor trained on one listening-test population does not reliably transfer to different conditions, which undermines comparisons against scores quoted from prior work. Running a separate listening panel for every system is also expensive; comparing three systems across two languages with matched statistical power would be difficult to do as a listening study, which is part of why we rely on automated metrics. The VoiceMOS Challenge [30] framed naturalness prediction as supervised regression over self-supervised speech representations, and UTMOS [10], the strongest submission to that challenge, has since become a common substitute for human MOS in benchmarking pipelines. It was trained predominantly on English and Japanese listening-test data, however, so its calibration on Hindi is not established, a caveat that applies to our own results as well.

Speaker similarity and intelligibility went through a similar shift. Where similarity was historically assessed with a dedicated similarity-MOS test, speaker-verification embeddings trained for a discriminative objective, from d-vectors [7] and x-vectors [31] to ECAPA-TDNN [8] and self-supervised WavLM [9], made cosine similarity between reference and synthesized embeddings a reproducible objective proxy. Intelligibility moved from manual transcription to automatic transcription once Whisper [11] made a single multilingual ASR model usable for WER estimation across languages without per-language fine-tuning. What has not followed is consistent adoption of all three proxies together. Individual papers tend to

report whichever subset suits the system and language at hand, and to our knowledge no prior work applies speaker similarity, ASR-derived error rate, and neural MOS to the same set of models in both a high-resource and an Indic language.

D. Positioning Relative to Prior Work and the Research Gap

The gaps described above line up along three axes. Architecturally, Orpheus-3B, VoxCPM2, and VibeVoice implement three distinct hypotheses about how speaker identity should be injected into a generative process (a token prefix in an autoregressive language model, cross-attention to a fixed prototype, and diffusion guided by a contrastive embedding), yet no published study places the three side by side under an identical reference-audio protocol. Linguistically, cross-lingual zero-shot cloning has been studied mainly among high-resource pairs [23] or as one arm of a broader multilingual benchmark [22], while Indic-focused research [24]–[26] has concentrated on recognition, leaving Hindi zero-shot cloning nearly absent as a dedicated comparative subject. Methodologically, the shift from subjective MOS to the automated triad of speaker-embedding similarity, ASR-derived WER/CER, and neural MOS has happened piecemeal, without a single pipeline that applies the same three metrics, computed from the same underlying models, across systems and languages. Without that, an observed difference may be an artifact of metric choice rather than of architecture.

The *Voice-Cloning-AiONOS* framework underlying this study is a response to that fragmentation. Rather than benchmarking one model on one language against whatever metric its original paper happened to report, it drives Orpheus-3B, VoxCPM2, and VibeVoice through a common inference interface and evaluates their output under the same 20–30 second reference protocol, the same target sentences, and the same implementations of speaker similarity, intelligibility, and naturalness, computed identically for English and Hindi. To our knowledge this is the first controlled multi-architecture, multi-metric comparison of zero-shot voice cloning that treats Hindi as a primary evaluation language rather than an incidental entry in a longer list.

III. METHODOLOGY

The evaluation is designed so that differences in the reported numbers can be attributed to the model under test rather than to the reference voice, the target text, the metric implementations, or the software that runs them. All of these are held fixed; only the synthesis system varies. The three systems are driven through a common inference interface (the *Voice-Cloning-AiONOS* harness introduced in Section II-D). Each receives the identical reference recording and the identical set of target sentences, produces one clip per sentence with no speaker-specific fine-tuning, and is scored by the same three metric implementations in both languages. Generation ran on a single GPU, with per-utterance synthesis times between roughly 16 and 65 seconds depending on the system and utterance length. Figure 1 shows the arrangement: one set of fixed inputs, three systems run at their released defaults, and one pool of

synthesized clips scored by three metric models that share no components with the systems under test.

A. Reference Voice and Target Corpus

All three systems clone the same reference recording of one speaker: 25.5 seconds of mono speech, inside the 20–30 second window that zero-shot systems are typically prompted with. The recording has a mean fundamental frequency of 118.8 Hz with a standard deviation of 13.9 Hz, a voiced-frame ratio of 0.66, and an estimated signal-to-noise ratio of 48 dB. Using one reference for every system removes speaker selection as a confound: any variation in measured identity preservation reflects what each system does with the same evidence, not differences in how much evidence it was given.

The target corpus is twelve short, goal-directed utterances of the kind produced by automated spoken-response systems, three in English and nine in Hindi. We chose utterances that stress a clone rather than ones that would maximize fluency scores. Several contain alphanumeric identifiers and spoken numbers (ticket-style codes, reference numbers, monetary amounts, dates), because these are the tokens on which an intelligibility metric is most discriminating: a single dropped or hallucinated digit changes the meaning while barely perturbing prosody. The Hindi items are written in romanized Latin script, with transliterated Hindi interleaved with English loanwords, which is how such prompts are usually authored in practice. That choice has direct consequences for how Hindi intelligibility has to be scored, discussed in Section III-C. Utterance durations after synthesis span roughly 4 to 15 seconds. The language split is deliberately asymmetric, nine Hindi items to three English, because English serves as the well-characterized control condition and Hindi is the object of study.

B. Systems Under Evaluation

Each system implements a different approach to speaker conditioning, as set out in Section II-A. We evaluate all three as publicly released, without architectural modification.

Orpheus-3B is a ~ 3 -billion-parameter language-model-based TTS system (the `orpheus-3b-0.1-pretrained` release). It treats speech as a sequence of discrete audio tokens and conditions cloning on a reference prefix, so identity is carried entirely by the backbone’s in-context learning and decoded through a neural audio codec.

VoxCPM2 is a ~ 2 -billion-parameter codec language model that replaces the prefix strategy with cross-attention to a fixed speaker representation extracted from the reference, an intermediate design between pure prompting and an explicit embedding bottleneck.

VibeVoice is a ~ 1.5 -billion-parameter diffusion-based system whose decoder is guided by a contrastively trained speaker encoder rather than an autoregressive codec, placing it in the representation-learning branch of the literature.

Every system received the same reference clip and target sentences and ran with its released default inference configuration. No per-speaker adaptation, per-sentence prompt engineering, or output cherry-picking was applied.

C. Evaluation Metrics

Each clip is scored on three complementary axes, plus a set of acoustic descriptors. Every metric model is independent of all three systems under test, so no system is scored by a component of its own pipeline.

Speaker similarity (SECS). We embed the reference and each synthesized clip with the Resemblyzer speaker encoder, a GE2E-style d-vector network [7] trained for speaker discrimination and unrelated to any of the TTS systems, and report the cosine similarity between the two 256-dimensional utterance embeddings. Both signals are resampled to 16 kHz mono before embedding. The absolute scale of such cosine scores is encoder-specific, so the SECS values here are meaningful for ranking the three systems under one encoder, but should not be compared directly against speaker-similarity numbers computed with a different encoder such as ECAPA-TDNN or WavLM.

Intelligibility (WER / CER). Every clip is transcribed with Whisper (the `small` model, beam size 5, with voice-activity filtering) [11] and compared against the input text. Both sides pass through a normalization stage before scoring: case folding, punctuation removal, mapping of spelled-out English numbers to digits, and merging of single-character alphanumeric runs so that a spoken code (“B 4 7 2 9”) and its written form (“B4729”) are scored as the same content rather than as a formatting difference. The Hindi measurement chain is necessarily longer, and we treat it as such. Whisper emits Hindi in Devanagari while our references are romanized, so we transliterate the hypothesis back to Latin script (Devanagari $\rightarrow$ IAST, the deterministic direction) and then apply a phonetic normalization that collapses romanization-convention noise: diacritics, vowel-length spellings, *w/v* alternation, and doubled consonants. We report **WER** as the primary intelligibility number for English and **CER** for Hindi. At the word level, schwa deletion and spelling variation push Hindi WER above 1.0 in many clips, whereas character-level comparison separates phoneme fidelity from tokenization artifacts. Because this pipeline depends on a multilingual ASR model of modest size and a rule-based romanization step, the Hindi error rates should be read as intelligibility *as measured under this ASR pipeline*, not as an ASR-independent ground truth.

Naturalness (UTMOS). Naturalness is scored with UTMOS [10], a non-intrusive neural mean-opinion-score predictor that maps a waveform to a 1–5 scale without a reference signal or a listening panel. UTMOS was trained predominantly on English and Japanese listening-test data, so its Hindi scores carry a calibration caveat that we return to when interpreting the cross-lingual results.

Acoustic descriptors. We also extract simple signal descriptors for each clip: fundamental-frequency mean and standard deviation via probabilistic YIN, the absolute pitch deviation from the reference ($|\Delta F_0|$), an SNR estimate, speaking rate, loudness, and voiced/silence ratios. Of these, $|\Delta F_0|$ is the most diagnostic for cloning, since a stable clone should

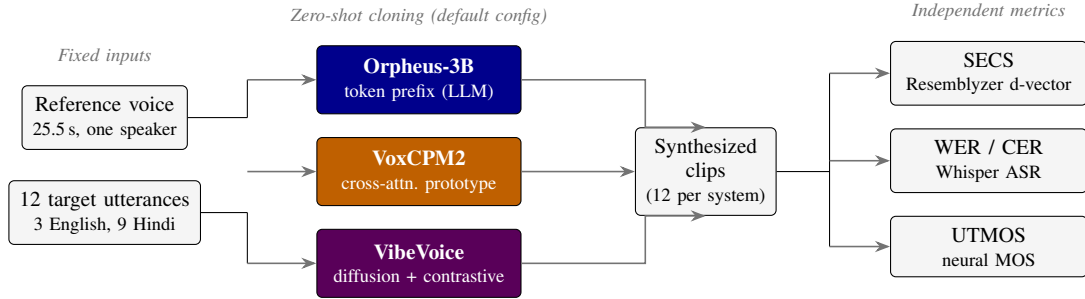


Fig. 1. The evaluation protocol. A single reference recording and one fixed set of target utterances drive all three systems, each run at its released default configuration with no per-speaker adaptation. Every synthesized clip is scored by the same three metric models, none of which shares components with any system under test, so differences in the reported numbers reflect the synthesis systems. System colours are reused in all later figures.

TABLE I

OVERALL SYSTEM COMPARISON, AVERAGED OVER ALL CLIPS EXCEPT WHERE A LANGUAGE IS SPECIFIED. ARROWS GIVE THE IMPROVING DIRECTION; **BOLD** MARKS THE BEST VALUE PER COLUMN. WER IS REPORTED FOR ENGLISH AND CER FOR HINDI (SEE SECTION III-C).

System	SECS ↑	UTMOS ↑	WER (En)↓	CER (Hi)↓	$ \Delta F_0 $ (Hz)↓
Orpheus-3B	0.894	3.51	0.116	0.548	17.7
VoxCPM2	0.948	3.73	0.079	0.416	2.5
VibeVoice	0.933	3.80	0.490	0.551	6.1

TABLE II

CROSS-LINGUAL BREAKDOWN (ENGLISH, $n = 3$; HINDI, $n = 9$). HINDI WER IS SHOWN IN PARENTHESES BUT IS NOT THE PRIMARY HINDI NUMBER, AS WORD-LEVEL SCORING OF ROMANIZED HINDI IS UNRELIABLE (VALUES FREQUENTLY EXCEED 1.0).

System	Lang.	SECS↑	UTMOS↑	WER↓	CER↓
Orpheus-3B	English	0.897	3.83	0.116	0.074
	Hindi	0.893	3.41	(0.983)	0.548
VoxCPM2	English	0.952	3.99	0.079	0.034
	Hindi	0.947	3.65	(0.857)	0.416
VibeVoice	English	0.930	3.90	0.490	0.267
	Hindi	0.934	3.77	(1.017)	0.551

reproduce the reference speaker’s register.

D. Aggregation and Scope

Metrics are computed per clip and then averaged within each system. Because the central question is cross-lingual, we report English and Hindi means separately rather than pooling them; word- and character-error rates in particular are not comparable across the two scripts. This is a controlled, small-scale comparison: one speaker, one reference, twelve utterances, and a single generation run per system. We therefore report descriptive results and effect sizes rather than significance tests, and read the numbers as evidence about these systems under this protocol rather than as a powered benchmark claim.

IV. RESULTS

Table I gives the headline comparison across all four measured dimensions, and Table II breaks the identity, intelligibility, and naturalness axes down by language. Figure 2 plots the three primary axes, Figure 3 the identity–naturalness trade-off by language, Figure 4 the per-clip spread behind the means, and Figure 6 pitch fidelity. No single system is best on every axis; the results are better described as a set of trade-offs than as an overall ranking.

A. Speaker Similarity

VoxCPM2 preserves speaker identity most faithfully, with a mean SECS of 0.948, followed by VibeVoice at 0.933 and Orpheus-3B at 0.894 (Table I, Fig. 2a). The gap between the best and worst system, 0.054 in cosine similarity, is the clearest separation on any identity-related axis. The per-clip scores

tell the same story: Figure 4a shows VoxCPM2’s twelve clips clustered tightly around its mean, while Orpheus-3B’s are both lower and more dispersed, ranging from about 0.85 to 0.93. Orpheus-3B is thus both the weakest and the least consistent of the three on identity.

Speaker similarity is almost unaffected by language within each system: the English and Hindi SECS means differ by at most 0.005 for any system (Table II, Fig. 5a). Whatever mechanism each architecture uses to carry identity from the reference clip transfers across the two scripts essentially intact, even though, as shown next, intelligibility does not. Identity preservation and linguistic fidelity appear largely decoupled in these systems.

B. Intelligibility

On English, where WER is directly interpretable, VoxCPM2 is again strongest (WER 0.079), Orpheus-3B is close behind (0.116), and VibeVoice is markedly higher (0.490). The VibeVoice figure is driven by two of its three English clips: one rendered a numeric ticket code as “B fold 729” and another drifted on a spoken date, while the third clip matched the best systems at WER 0.036. With only three English items, we report this as an observation about specific numeric and alphanumeric utterances rather than as a distributional claim, though it does match the corpus design intent that spoken codes are where intelligibility diverges.

The dominant effect in the data is the cross-lingual gap. Hindi character error rates are far higher than English word

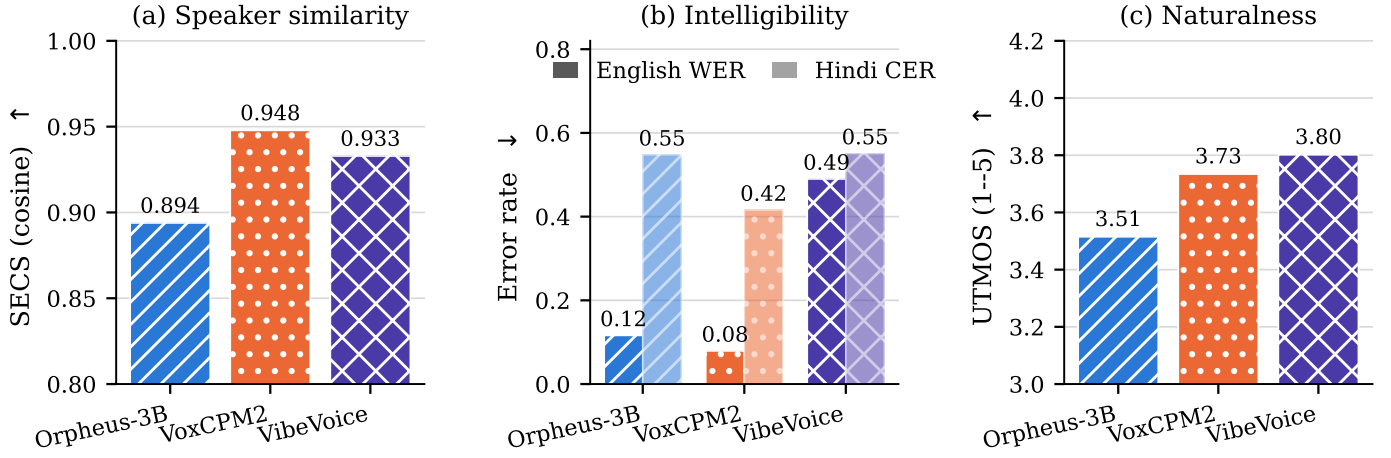


Fig. 2. The three primary evaluation axes. (a) Speaker similarity (SECS, cosine, all clips); (b) intelligibility, contrasting English WER with Hindi CER; (c) naturalness (UTMOS). Each system keeps the same colour and hatch in all figures.

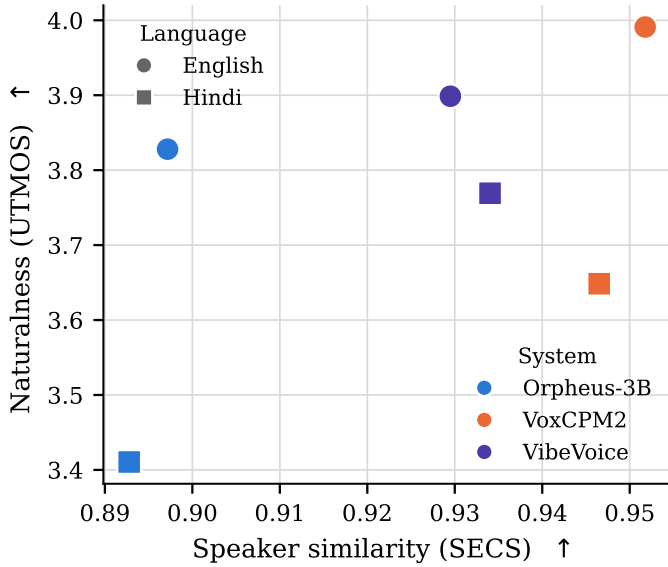


Fig. 3. The identity–naturalness trade-off. Each point is a system’s mean over one language (circles: English; squares: Hindi). VoxCPM2 sits at the high-similarity edge, VibeVoice at the high-naturalness edge, and Orpheus-3B is dominated on both.

error rates for every system (VoxCPM2 0.416, Orpheus-3B 0.548, VibeVoice 0.551; Fig. 2b), and character error rates alone rise roughly four- to eight-fold from English to Hindi (Table II). Two causes are entangled here, and we are careful not to attribute the whole gap to synthesis. Part of it is genuine difficulty: Hindi’s retroflex consonants, schwa deletion, and heavy loanword content are harder to render cleanly, and the per-clip spread in Figure 4b is correspondingly wide. Part of it is measurement: the Hindi score passes through a modest multilingual ASR model and a rule-based Devanagari-to-Latin romanization, and both inject error that inflates CER independently of audio quality. The ranking among systems is consistent enough to be informative, with VoxCPM2 leading on Hindi as it does on English, but the absolute Hindi numbers

should be read with the pipeline of Section III-C in mind.

C. Naturalness

Predicted naturalness is the axis on which the three systems are most alike. UTMOS means fall in a narrow band from 3.51 (Orpheus-3B) to 3.80 (VibeVoice), with VoxCPM2 between them at 3.73 (Fig. 2c). VibeVoice, the diffusion-based system, scores highest on this predictor and is also the most stable across languages, losing only 0.13 UTMOS from English to Hindi, whereas Orpheus-3B and VoxCPM2 drop by about 0.4 and 0.3 respectively (Table II, Fig. 5b). Since UTMOS is not calibrated on Hindi, we read the Hindi values as within-study relative indicators rather than as absolute mean-opinion scores. Figure 5 places the two axes side by side: the speaker-similarity lines stay nearly flat across the language boundary while the naturalness lines slope downward. This is the clearest single view of the identity/content split that Section V builds on.

D. Pitch Fidelity and the Overall Trade-off

The acoustic descriptors support the identity ranking. VoxCPM2 reproduces the reference speaker’s register almost exactly, with a mean fundamental-frequency deviation of just 2.5 Hz, against 6.1 Hz for VibeVoice and 17.7 Hz for Orpheus-3B (Fig. 6). Since $|\Delta F_0|$ is an absolute quantity, it hides the direction of the error, and the direction is itself informative. Figure 7 plots the raw per-clip pitch against the reference: Orpheus-3B does not scatter symmetrically around it but sits above it on every clip, while VoxCPM2 clusters tightly on the reference line and VibeVoice straddles it more loosely. This register shift lines up with Orpheus-3B’s lower and more variable SECS. The system that departs most from the reference pitch is also the one that scores lowest on embedding-based identity.

Taken together (Fig. 3), the systems occupy distinct positions rather than a single ordering. VoxCPM2 is best or tied-best on speaker similarity, on English and Hindi intelligibility, and on pitch fidelity, at a small cost in predicted naturalness.

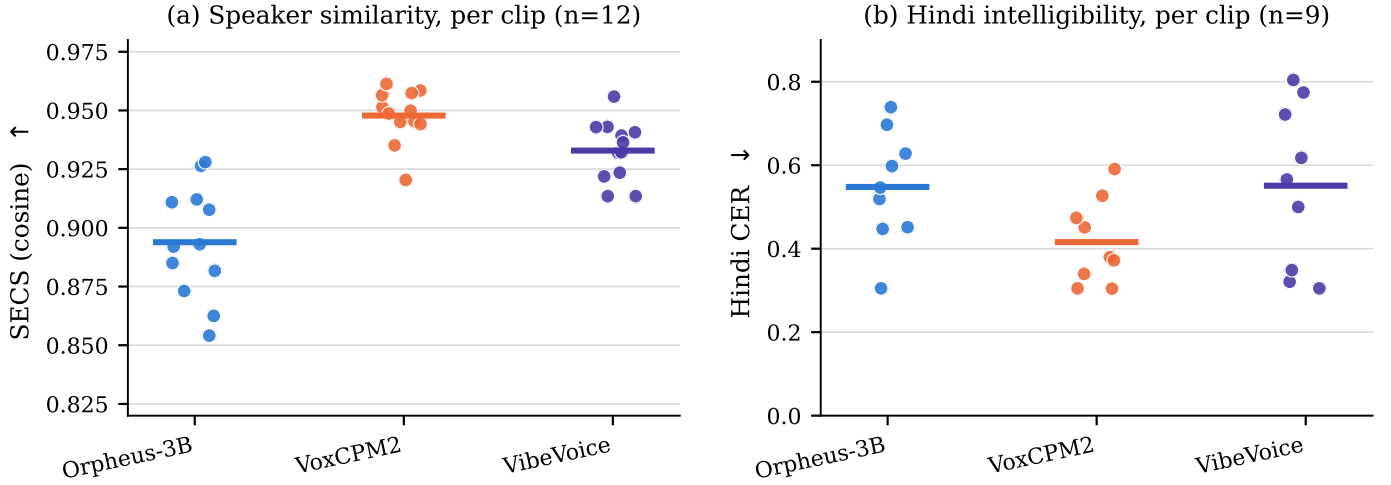


Fig. 4. Per-clip spread behind the means. Horizontal bars mark the per-system mean. (a) SECS over all 12 clips; VoxCPM2 is both highest and tightest. (b) Hindi CER over the 9 Hindi clips; VoxCPM2 is lowest, and every system shows substantial clip-to-clip variation.

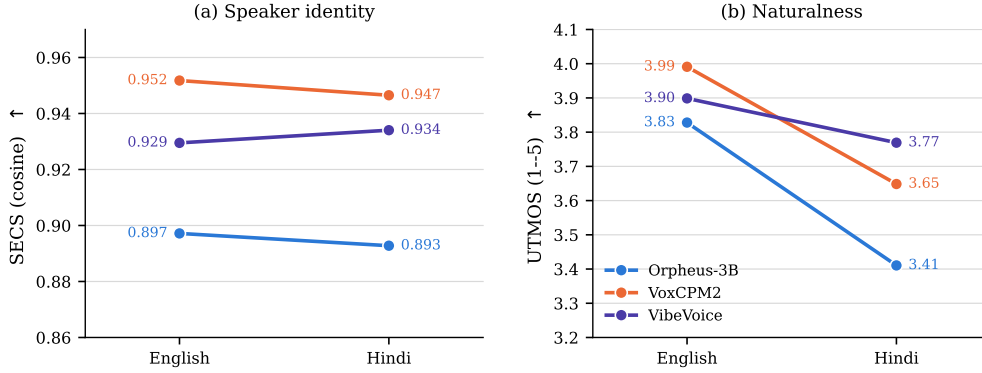


Fig. 5. Cross-lingual stability, English versus Hindi, one line per system. (a) Speaker similarity barely moves across the language boundary (at most 0.005 within any system); (b) naturalness drops for all systems but by different amounts: 0.13 UTMOS for VibeVoice against about 0.34 and 0.42 for VoxCPM2 and Orpheus-3B.

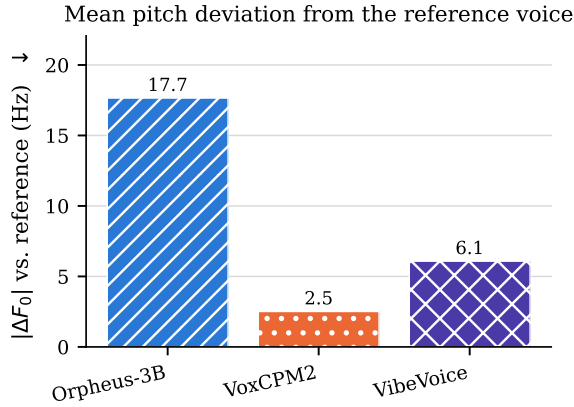


Fig. 6. Mean absolute deviation of synthesized pitch from the reference voice's fundamental frequency. VoxCPM2 tracks the reference register almost exactly, while Orpheus-3B shifts it by nearly 18 Hz.

VibeVoice is the most natural and the most language-stable, but has the weakest intelligibility. Orpheus-3B is dominated

on the identity and pitch axes and sits in the middle on intelligibility and naturalness. Section V relates these positions back to the three conditioning mechanisms described in Section I.

V. DISCUSSION

The results in Section IV separate into two largely independent stories: how well a system holds onto the speaker, and how faithfully it renders the words. A system can do well on one and poorly on the other, and the data show exactly that.

Identity behaves in a way we did not anticipate. Within every system, the English and Hindi SECS means sit within 0.005 of each other (Table II). Whatever each architecture does to carry the speaker from the reference clip into new speech works about as well in a language the reference speaker never used, at least for this speaker and this encoder. Intelligibility shows the opposite pattern: every system loses heavily moving from English to Hindi, with character error rates rising roughly four- to eight-fold.

This dissociation constrains how the Hindi error rates should be read. If a system were genuinely losing its grip on the

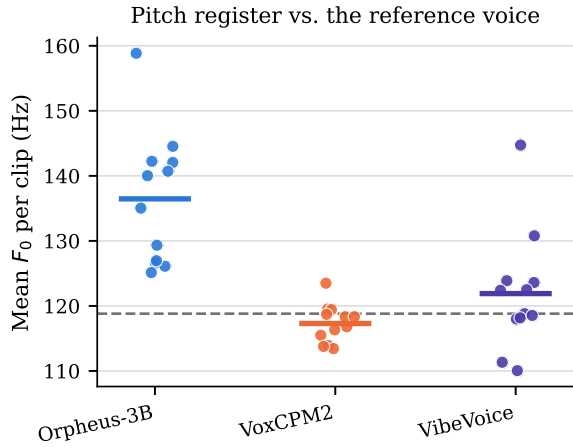


Fig. 7. Per-clip mean fundamental frequency for each system against the reference voice (dashed line, 119 Hz); horizontal bars mark per-system means. Unlike the absolute $|\Delta F_0|$ of Fig. 6, this view shows direction: Orpheus-3B raises the register on every clip, VoxCPM2 tracks the reference, and VibeVoice varies around it.

speaker when it switched to Hindi, we would expect the identity embedding to move as well, and it does not. The speaker-conditioning path and the content-generation path fail independently, which suggests that a substantial part of the Hindi CER gap is measurement rather than synthesis: the Whisper-small transcription and the rule-based Devanagari-to-Latin step (Section III-C) both introduce errors that have nothing to do with whether the voice sounds like the reference. SECS, scored by a pipeline that never touches the ASR chain, stays flat. We take this as the clearest sign in our data that the two axes measure different things and fail for different reasons.

The identity axis also has a concrete acoustic correlate, which is reassuring given how abstract a cosine distance in embedding space is. The pitch-deviation ranking matches the SECS ranking exactly. VoxCPM2 tracks the reference register to within 2.5 Hz and takes the top identity score; Orpheus-3B shifts the register by nearly 18 Hz and takes the bottom one; VibeVoice sits in the middle on both. The Orpheus-3B shift is not random: it raises pitch on every clip (Fig. 7), so the drift is a systematic property of the system rather than clip-level noise, and the d-vector embedding appears sensitive to it. A listener who perceived Orpheus-3B as pitched up relative to the reference and the embedding that scores it lower would be reacting to the same property.

The conditioning mechanisms described in Section I offer a plausible explanation for the identity ranking, though with three systems we can only note that the results are consistent with it, not prove it. Orpheus-3B carries the speaker as a token prefix and then generates autoregressively, so the only route back to the reference is the backbone’s in-context memory, which has to survive a long decoding rollout with no dedicated pathway to the original evidence. VoxCPM2 keeps a fixed speaker representation and cross-attends to it at every step, so the reference remains available late in generation rather than

only at the start. That VoxCPM2 is both highest and tightest on SECS while Orpheus-3B is lowest and most dispersed (Fig. 4a) is what one would expect if persistent access to the speaker matters more than the raw capacity of the backbone holding it in context.

VibeVoice illustrates a different trade-off. Its diffusion decoder produces the highest UTMOS and the most language-stable naturalness, losing only 0.13 crossing into Hindi against roughly 0.3 to 0.4 for the two codec-based systems. At the same time it has the weakest intelligibility of the three, in both languages: an English WER of 0.490 and the highest Hindi CER at 0.551. The English figure rests on a very small sample, since two of its three English clips carried numeric or alphanumeric content and both stumbled on it while the third matched the best systems, so we do not put weight on its magnitude. The Hindi CER, however, supports the same ranking on separate data. The overall pattern fits what a diffusion decoder optimizes for: a smooth, natural waveform rather than the exact discrete tokens that spell out a code or a date. Rendering a digit correctly is an all-or-nothing target, and a codec language model with a token-level objective appears to handle it more reliably than a decoder operating over continuous acoustic trajectories.

None of this yields a single best system, and forcing one would misrepresent the data. VoxCPM2 is the reasonable default when identity and register fidelity are the priority: it wins or ties on speaker similarity, on English and Hindi intelligibility, and on pitch, and it pays only a small and arguably inaudible price in predicted naturalness. VibeVoice is the better choice where fluent, natural delivery matters more than verbatim accuracy and where cross-lingual stability is valued, provided the text is not dense with codes and numbers. Orpheus-3B, on this protocol, is dominated on identity and pitch and roughly average elsewhere; its LLM-prefix design is conceptually the simplest of the three but, in practice, the loosest at holding a voice.

These readings carry the limits the design imposes. The comparison rests on one speaker, one reference clip, and twelve utterances scored in a single generation run, so we report effect sizes and describe trade-offs rather than claim statistical significance (Section III-D). The English intelligibility numbers in particular rest on three clips and should be read as observations about specific numeric utterances. UTMOS is uncalibrated on Hindi, so the Hindi naturalness scores are within-study indicators, not absolute opinion scores. And the Hindi error rates include whatever the ASR-and-romanization chain contributes on top of genuine synthesis error; the dissociation argument above suggests that contribution is substantial, but it does not let us subtract it cleanly. What the study does support is narrower and, we think, still useful: on matched inputs and matched metrics, persistent cross-attention to a speaker prototype held identity better than an autoregressive prefix, a diffusion decoder traded token-level accuracy for naturalness, and speaker identity transferred across scripts far more readily than linguistic content did.

VI. CONCLUSION

We compared three zero-shot voice cloning systems, Orpheus-3B, VoxCPM2, and VibeVoice, under a single controlled protocol: the same reference recording, the same twelve English and Hindi utterances, and the same speaker-similarity, intelligibility, and naturalness pipelines for every system. No system won on every axis. VoxCPM2 held the speaker’s identity and register most tightly and was the most intelligible in both languages. VibeVoice scored highest on naturalness and changed least across the language boundary. Orpheus-3B trailed on identity and pitch without leading anywhere else.

Three findings seem robust despite the study’s small scale. Persistent cross-attention to a fixed speaker prototype preserved identity better than conditioning on an autoregressive prefix, and the difference had an audible correlate in how faithfully each system tracked the reference pitch. The diffusion decoder produced the most natural and most language-stable speech but was the least reliable at rendering exact content such as digits and codes. And speaker identity crossed from English into Hindi almost unchanged while intelligibility did not, a dissociation that separates the two failure modes and indicates that much of the measured Hindi error belongs to the ASR-and-romanization scoring chain rather than to the synthesis itself.

Future work follows from the limitations. The single-speaker, twelve-utterance design should be widened to many speakers and references, with enough items per language to support significance testing rather than effect sizes. The naturalness and intelligibility proxies need grounding against human listeners, especially for Hindi, where UTMOS is uncalibrated and the CER pipeline does measurable work of its own; a Hindi-native ASR model and evaluation targets written in Devanagari would help separate synthesis quality from transcription noise. Broadening the language set beyond a single Indic language would show whether the ranking of conditioning mechanisms observed here is a general property or an artifact of this particular pair. Finally, we would like to probe the mechanisms directly by swapping prefix conditioning for cross-attention within one architecture, holding everything else fixed, to test the claim that persistent access to the speaker, rather than the size of the model holding it, is what keeps a cloned voice in register.

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